SupplyShock AI — disruption report

Port congestion · generated 25/10/2026, 6:06:15 pm · demo network: regional medical-supply distributor

What happened

Bayport Terminal can’t berth ships for about 4 days (storm backlog at the terminal). It starts Day 1 · 06:30.

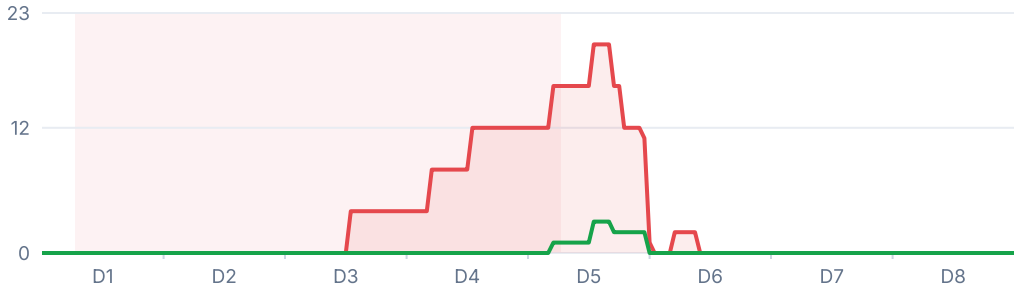
Impact if nothing is done

Late orders	22
Hospital orders late	11
Back to normal after	5.1 days
Late orders in a bad week (1 in 10 of 40 futures)	37

- Day 1 · 06:30 — Port congestion
- Day 2 · 14:00 — Central Hub runs out of insulin
- Day 2 · 14:00 — South Hub runs out of insulin
- Day 3 · 13:00 — St. Clara Hospital gets a late order
- Day 3 · 13:00 — Riverside Hospital gets a late order
- Day 3 · 13:00 — Harbor Pharmacy gets a late order
- Day 3 · 13:00 — Maple Care Home gets a late order

Recovery plan: Backup supplier + Rebalance stock + Hospitals first

- **Rush order from Ridgeway Labs.** 3,560 units in 8 batches, first trucks out 8 h after ordering · +40% unit cost
- **Move stock to where it’s needed.** North → South 432 · North → Central 168 units
- **Hospitals first.** Hospital orders ship first and half a day of their demand is held back, until a day after the disruption ends



Red: orders overdue with no action. Green: with the plan.

How the options compare

	Do nothing	Chosen plan Backup supplier + Rebalance stock + Hospitals first	Runner-up Backup supplier + Hospitals first
Late orders	22	3	8
Hospital orders late	11	0	0
Back to normal after	5.1 days	4.7 days	4.7 days
Disruption cost	$64k	$28k	$27k
Extra CO_2	≈0 t	5.0 t	2.2 t
Hospitals safe in (of 40 futures)	0%	100%	100%

Rows show the expected course of events; the last row shows how often every hospital order arrived on time when demand, travel times and the disruption's length were varied.

Recommendations

- Without action the first stockout hits Central Hub (insulin) at Day 2 · 14:00 — 1.3 days after the disruption starts.
- On its own, "Backup supplier" was the strongest single move: 3 late orders instead of 22.
- Pre-negotiate a standing rush agreement with a local backup supplier so it can start within hours.
- North Hub's extra safety stock acted as the network's shock absorber; keep a regional reserve like it.
- Agree a hospital-first rationing rule in advance so scarce stock goes to patients automatically.

Method

328 discrete-event simulations (8 strategies × 40 futures + expected case). Each future varies order sizes (±15%), truck travel times (±8%) and the disruption's real length (−20% to +35%). Plans are ranked with late hospital orders weighted highest, then the chosen objective. Customer and site names are fictional.